

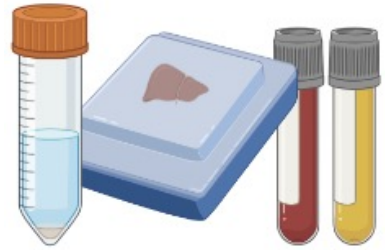
AMIST 인턴십 활동 보고

Clinical Proteomics Core Lab.

1. Sample preparation
2. Instrumental analysis
3. Bioinformatics
4. BioFD&C_KEH

1. Sample preparation

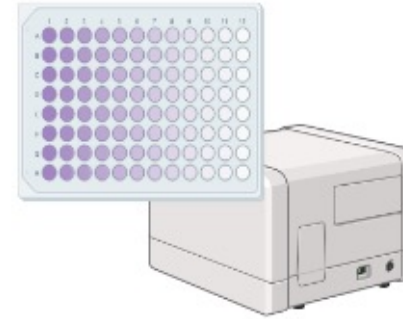
– overview of sample preparation protocol



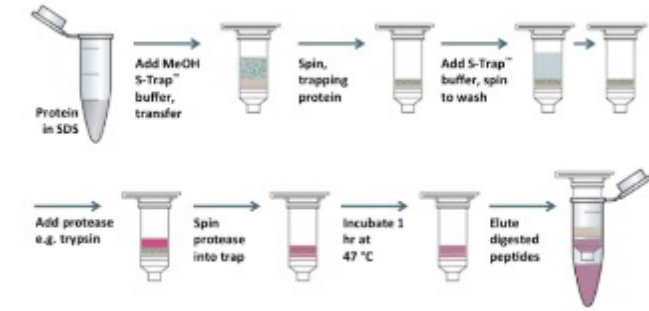
Sample



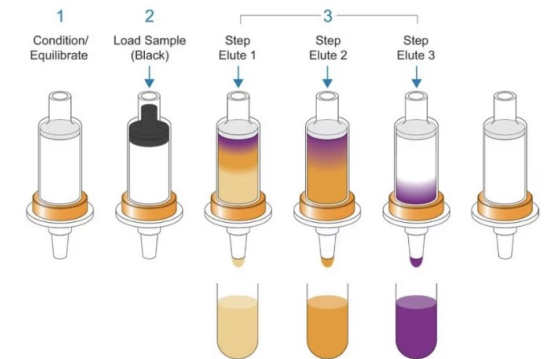
Lysis



Quantification of proteins



Digestion of proteins



Desalting



Quantitation of peptides



Dry & Ready



Reconstitution
&
Vial transfer

1. Sample preparation

– Sample lysis



Cell



Probe sonication

1. Sample preparation

– Sample lysis



Blood



centrifugation

1. Sample preparation

– Sample lysis



FFPE

(formalin fixed paraffin embedded)



AFA(adaptive focused acoustics)
sonication

1. Sample preparation

– Quantification of proteins

Bicinchoninate assay (BCA assay)



- BCA assay 원리

단백질이 구리 이온을 환원 시킬 수 있는 성질을 이용한 흡광도 측정 방법

BCA assay reagent A



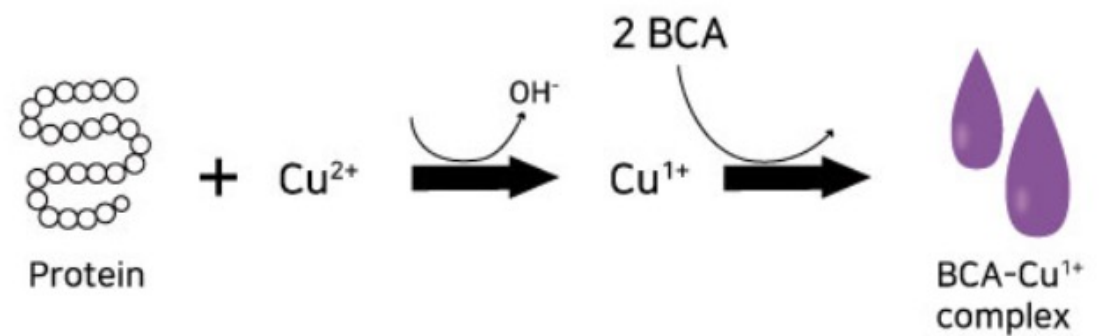
50

BCA assay reagent B



1

:

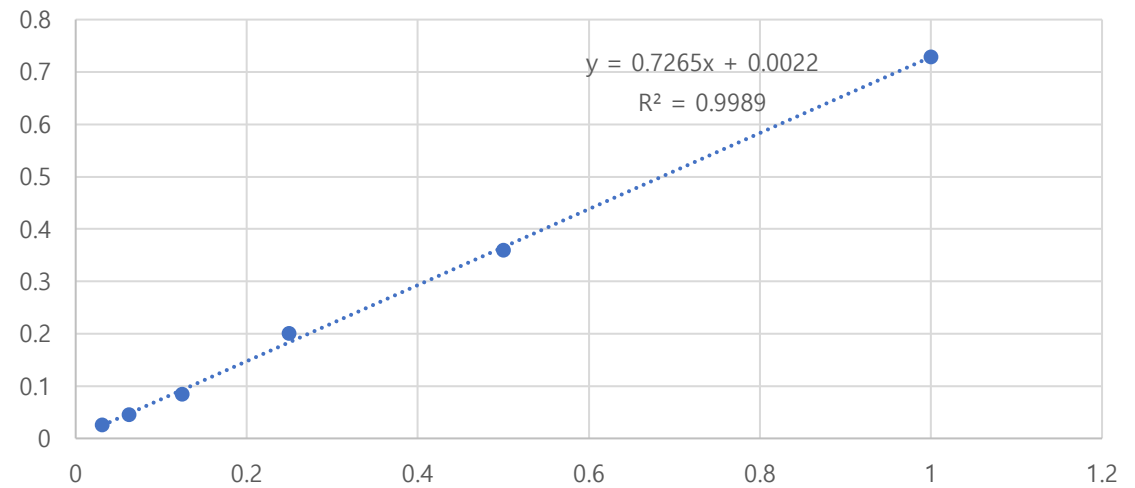


1. Sample preparation

– Quantification of proteins

STD conc. ($\mu\text{g}/\mu\text{l}$)		Absorbance (562nm) =OD-Blank
1	1.1024	0.73
0.5	0.7330	0.36
0.25	0.5736	0.20
0.125	0.4577	0.08
0.0625	0.4188	0.05
0.03125	0.3993	0.03
BL	0.3736	0.00

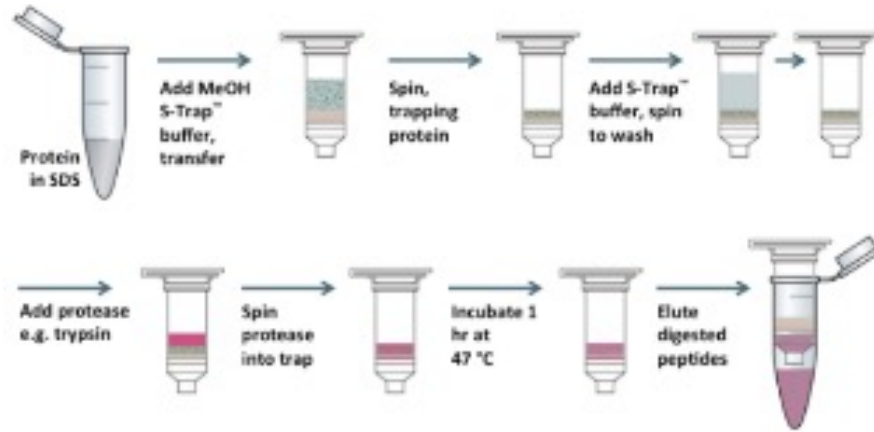
BCA assay Standard Absorbance(562nm) Curve



1. Sample preparation

– Digestion of proteins

Suspension trapping (S-Trap)



- ① Lysis buffer
 - ② Reduction : DTT*
 - ③ Denaturation : Heat at 95°C
 - ④ Alkylation : IAA* (DTT의 2배)
 - ⑤ Incubate in dark
 - ⑥ Phosphoric acid
 - ⑦ Binding buffer → S-trap tube
 - ⑧ Centrifuge
 - ⑨ Wash (3~5 times)
 - ⑩ Trypsin
 - ⑪ Incubate for overnight(16hrs)
- (Elution)
- ① TEAB
 - ② 0.2% FA
 - ③ 0.2% FA in 50% ACN

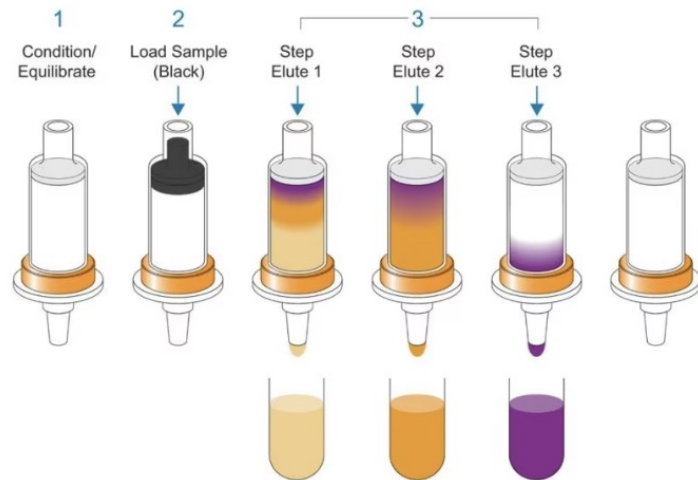
*DTT (dithiothreitol) – disulfide bond 끊어줌

*IAA (iodoacetamide) – 끊어진 disulfide bond 재결합 방지

1. Sample preparation

– Desalting

SPE (Solid Phase Extraction)



- SPE의 필요성
: 분석기기 성능의 향상, 시료에 포함된 불순물 제거

- SPE 단계 요약
- ① ACN 800ug *2 – filter activation
 - ② 0.1FA 800ug *2 – equilibration
 - ③ Sample binding
 - ④ 0.1FA 800ug*2 – washing
 - ⑤ 40% ACN in 0.1FA 300ug *2 – elution

1. Sample preparation

– Quantitation of peptides / Dry and Ready

Nanodrop



speedvac



Peptide 정량 값을 토대로 sample간
농도를 맞춰주기 위해 dry



0.1% FA로 reconstitution



사용할 기기 종류에 알맞게
vial transfer

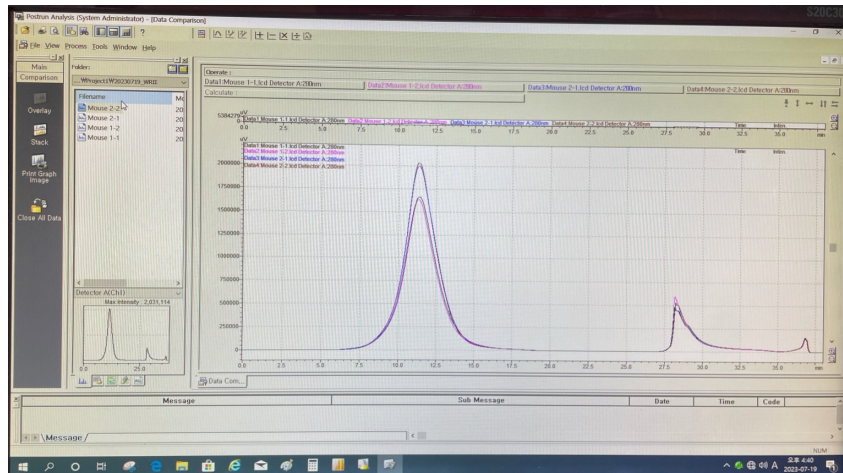


Queue up

2. Instrumental analysis

– Liquid chromatography

HPLC(High-Performance Liquid Chromatography)



LC(Liquid Chromatography)



2. Instrumental analysis

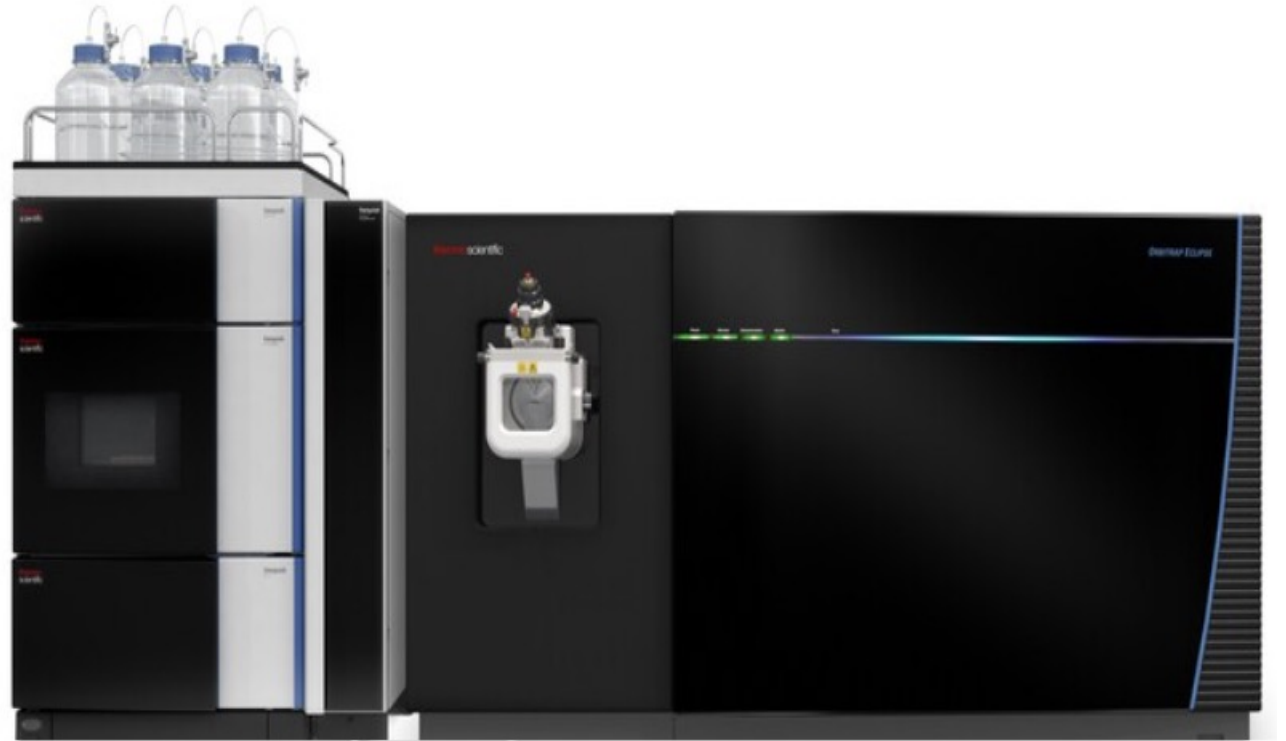
– Mass spectrometer

**HF-X Hybrid Quadrupole-Orbitrap
Mass Spectrometer**



OT-OT

**Orbitrap Eclipse
Mass Spectrometer**



OT-OT
IT-OT

3. Bioinformatics

– Perseus

Perseus

: 대량의 단백질-단백질 상호작용 및 다양한 실험 데이터를 분석하고 시각화하는 software

랩실 데이터

CADASIL raw data



3. Bioinformatics

– GO (Gene Ontology)

GO(Gene Ontology)

: 유전자 기능 연구를 위해 개별 유전자(gene)에 대해 유전자가 관련된 세포 기작(biological process), 분자기능(molecular functions), 세포 내의 위치(cellular component)를 주석(annotation)으로 표현하는 구조화된 모델

DAVID

ShinyGO

Enrichr

3. Bioinformatics

– GO (Gene Ontology)

1) DAVID

- Database for Annotation, Visualization and Integrated Discovery의 약자로 유전체 데이터를 기반으로 기능적 주석, 시각화, 통합 분석을 제공하는 데이터베이스
- 주로 유전자 및 단백질 함수, 경로 및 생물학적 과정을 이해하는 데 사용됩니다.

DAVID Bioinformatics Resources 6.8
Laboratory of Human Retrovirology and Immunoinformatics (LHRI)

Home | Start Analysis | Shortcut to DAVID Tools | Technical Center | Downloads & APIs | Term of Service | Why DAVID? | About Us

*** Welcome to DAVID 6.8 ***
*** If you are looking for DAVID 6.7, please visit our development site. ***

Shortcut to DAVID Tools

- Functional Annotation**
Gene-annotation enrichment analysis, functional annotation clustering, BioCarta & KEGG pathway mapping, gene-disease association, homologue match, ID translation, literature match and more.
- Gene Functional Classification**
Provide a rapid means to reduce large lists of genes into functionally related groups of genes to help unravel the biological context captured by high throughput technologies. [More](#)
- Gene ID Conversion**
Convert list of gene ID/accessions to others of your choice with the most comprehensive gene ID mapping repository. The ambiguous accessions in the list can also be determined semi-automatically. [More](#)
- Gene Name Batch Viewer**
Display gene names for a given gene list. Search functionally related genes within your list or not in your list. Deep links to enriched detailed information. [More](#)

Welcome to DAVID 6.8

2003 - 2018

The Database for Annotation, Visualization and Integrated Discovery (DAVID) v6.8 comprises a full Knowledgebase update to the sixth version of our original web-accessible programs. DAVID now provides a comprehensive set of functional annotation tools for investigators to understand biological meaning behind large list of genes. For any given gene list, DAVID tools are able to:

- Identify enriched biological themes, particularly GO terms
- Discover enriched functional-related gene groups
- Cluster redundant annotation terms
- Visualize genes on BioCarta & KEGG pathway maps
- Display related many-genes-to-many-terms on 2-D view.
- Search for other functionally related genes not in the list
- List interacting proteins
- Explore gene names in batch
- Link gene-disease associations
- Highlight protein functional domains and motifs
- Redirect to related literatures

What's Important in DAVID?

- Cite DAVID
- IDs of Affy Exon and Gene arrays supported
- Novel Classification Algorithms
- Pre-built Affymetrix and Illumina backgrounds
- User's customized gene background
- Enhanced calculating speed

Statistics of DAVID

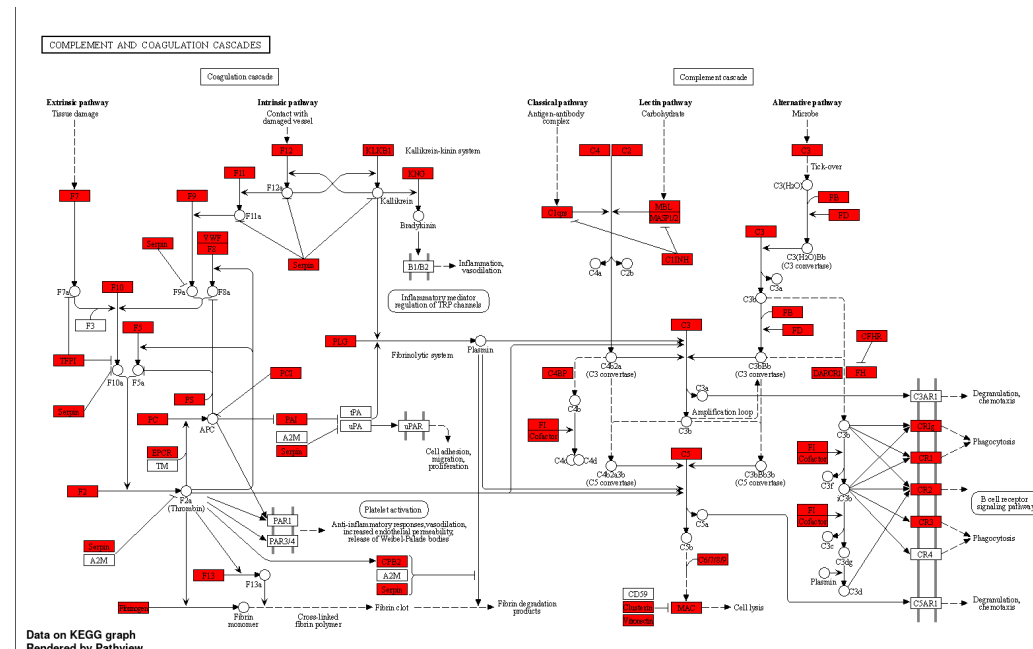
DAVID Citations (2003-2017)

Year	Citations
03	100
04	200
05	300
06	400
07	500
08	600
09	700
10	800
11	900
12	1000
13	1100
14	1200
15	1300
16	1400
17	1500

- > 33,000 Citations
- Average Daily Usage: ~2,700 gene lists/sublists from ~900 unique researchers.

- GO (Gene Ontology)

- 생물학적 용어의 기능적 해석 및 유전자 및 단백질의 기능을 이해하기 위한 tool
- 시각화된 네트워크를 통해 상호작용과 기능적 연결을 시각화
- 비록 그 종 범위가 DAVID만큼 광범위하지는 않지만, ShinyGO는 인간, 쥐 등의 miRNA 표적 유전자에 관한 더 포괄적인 유전자 set을 가지고 있습니다.



3. Bioinformatics

– GO (Gene Ontology)

3) Enrichr

Enrichr를 통해 사용자는 인간 또는 마우스 유전자 목록을 제출하여 전사 인자에 의해 조절되는 경로, 질병 또는 유전자 세트와 같은 알려진 생물학적 기능의 수많은 유전자 세트 라이브러리와 비교할 수 있습니다.

The screenshot displays the Enrichr website interface. At the top, the Enrichr logo is visible, along with navigation links for Transcription, Pathways, Ontologies, Diseases/Drugs, Cell Types, Misc, Legacy, and Crowd. A 'Login | Register' link is also present. Below the navigation bar, a search bar contains the text 'Top 100 genes co-express with STAT3 identified with ARCHS4 RNA-seq gene-gene co-expression matrix (100 genes)'. The search results are organized into a grid of 12 panels, each representing a different database or pathway collection. Each panel lists the top enriched terms and their associated p-values.

Database/Pathway	Top Enriched Terms
Reactome 2022	Cytokine Signaling In Immune System R-HSA-168256, Immune System R-HSA-168256, Interferon Signaling R-HSA-913531, Interferon Gamma Signaling R-HSA-877300, Signaling By Interleukins R-HSA-449147
BioPlanet 2019	Immune system, Immune system signaling by interferons, int, Interleukin-2 signaling pathway, T cell receptor regulation of apoptosis, Interferon signaling
WikiPathway 2021 Human	Type II Interferon signaling (IFNG) WP619, Immune response to tuberculosis WP4197, Apoptosis WP254, IL-4 signaling pathway WP395, Apoptosis Modulation and Signaling WP177
KEGG 2021 Human	NOD-like receptor signaling pathway, TNF signaling pathway, Necroptosis, Influenza A, JAK-STAT signaling pathway
ARCHS4 Kinases Coexp	RIPK1 human kinase ARCHS4 coexpression, JAK2 human kinase ARCHS4 coexpression, MLKL human kinase ARCHS4 coexpression, RIPK3 human kinase ARCHS4 coexpression, ERN1 human kinase ARCHS4 coexpression
Elsevier Pathway Collection	IFN-gamma/TNF-alpha Mediated Cell Proliferation, beta-Cell Destruction in Diabetes Mellitus Type 2, CD4+T-Cell Response in Celiac Disease, Natural Killer Cell Activation, T-Cell Cytotoxicity Mediated Cell Death
MSigDB Hallmark 2020	Interferon Gamma Response, Interferon Alpha Response, IL-6/JAK/STAT3 Signaling, Allograft Rejection, TNF-alpha Signaling via NF-kB
BioCarta 2016	IL22 Soluble Receptor Signaling Pathway Homo sapiens, Antigen Processing and Presentation Homo sapiens, IFN gamma signaling pathway Homo sapiens, IFN alpha signaling pathway Homo sapiens, TNFR2 Signaling Pathway Homo sapiens
HumanCyc 2016	protein ubiquitylation Homo sapiens PWY-7

3. Bioinformatics

– singscore

singscore

- ▶ 0. data 업로드
- ▶ 1. singscore 설치
- ▶ 2. sample scoring
- ▶ 3. sample scoring with a reduced number of measurements
- ▶ 4. visualisation and diagnostic functions
- ▶ 5. estimate empirical p-values for the obtained scores in individual samples and plot null distributions

3. Bioinformatics

– singscore

singscore

▼ 0. data 업로드

1) csv 파일(.csv)

```
install.packages("readr")
library(readr)

myData <- read.csv("data 위치")
View(myData)
```

2) excel 파일(.xls or .xlsx)

```
install.packages('readxl')
library(readxl)

myData <- read.excel("data 위치")
View(myData)
```

만약, 다음과 같은 error가 발생한다면

```
Error in check_for_XQuartz(file.path(R.home("modules"), "R_de.so")) :
X11 library is missing: install XQuartz from www.xquartz.org
```

www.xquartz.org에서 XQuartz를 설치한 후 R을 다시 실행하면 error를 해결할 수 있습니다.

▼ 1. singscore 설치

- 'singscore' from Bioconductor

```
if (!requireNamespace("BiocManager", quietly=TRUE))
  install.packages("BiocManager")
BiocManager::install("singscore")
```

- 가장 최신 버전의 'singscore'

```
install.packages('devtools')
devtools::install_github('DavisLaboratory/singscore')
```

▼ 2. sample scoring

1. `rankGenes()` 함수: 유전자 발현 data set을 입력으로 받아 rank matrix를 반환합니다. 즉, 각 유전자들을 sample 내에서의 상대적인 발현 수준에 따라 순위를 매긴 matrix를 생성합니다.
2. `simpleScore()` 함수: `rankGenes()` 함수로 얻은 랭크 매트릭스와 기준 유전자 세트(signatures)를 입력으로 받습니다. `upSet` 과 `downSet` 매개변수를 통해 상향 및 하향 조절된 유전자 세트를 지정합니다.
 - `upSet`: 상향 조절된 유전자들의 유전자 ID가 들어있는 벡터 또는 리스트입니다.
 - `downSet`: 하향 조절된 유전자들의 유전자 ID가 들어있는 벡터 또는 리스트입니다.

⇒ 이 function은 각 sample에 대해 상향 조절된 유전자들과 하향 조절된 유전자들에 대한 점수를 계산하고, 이를 합산한 `TotalScore` 와 함께 결과를 반환합니다. 또한, 각각의 점수와 함께 `dispersions` (산포도)도 계산하여 반환합니다.

▼ 3. sample scoring with a reduced number of measurements

- 안정적으로 발현되는 일부 유전자를 활용하여 상대적인 유전자 발현을 평가
- 패널 기반의 유전체 분석 방법에서 작은 유전자 패널로 작업할 수 있도록 해주는 방법

1. stable 유전자 얻기

- 코드는 `getStableGenes()` 함수를 사용하여 stable하게 발현되는 유전자 목록을 얻습니다.